

Comparative Performance and Risk Analysis of NASDAQ-100 and S&P 500

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Table Of Contents

1. Executive Summary	2
2. Business Problem & Analytical Objective	2
3. Data Overview.....	3
3.1 Overview of QQQ and SPY	3
3.2 Time Period and Data Granularity.....	3
3.3 Key Data Fields Used for Analysis	3
4. Tools & Technologies Used	4
5. Data Modeling in Power BI.....	4
5.1 Star Schema Design	4
5.2 Fact Returns Table.....	5
5.3 Metrics Table.....	5
5.4 Date Dimension and Relationships	5
5.5 Rationale for the Chosen Model.....	5
6. Dashboard Design Philosophy	6
7. Dashboard Page 1: Executive Overview	6
7.1 Year Slicer	7
7.2 CAGR – QQQ.....	7
7.3 CAGR – SPY	7
7.4 Cumulative Return Line Chart	8
7.5 Key Insight from Page 1.....	8
8. Dashboard Page 2: Risk Analysis	9
8.1 Risk Metrics Summary Table	9
8.2 Annualized Volatility – QQQ vs SPY	10
8.3 Maximum Drawdown – Downside Risk.....	10
8.4 Sharpe Ratio – Risk Adjusted Return.....	11
8.5 Key Insight from Risk Analysis	11
9. Dashboard Page 3: Performance Breakdown.....	12
9.1 Rolling 12-Month Return – QQQ vs SPY	12
9.2 Daily Return Distribution – QQQ vs SPY	13
9.3 Best vs Worst Monthly Returns Table	13
9.4 Key Insights from Performance Breakdown	14
10. Key Findings & Investment Interpretation	14
11. Limitations & Assumptions	14
12. Conclusion	15

1. Executive Summary

This project presents a comparative performance and risk analysis of two major U.S. equity exchange-traded funds (ETFs): QQQ and SPY, implemented through an interactive Power BI dashboard and supported by a Python-based data pipeline.

The objective of the analysis is to evaluate whether the higher returns of the NASDAQ-100 (QQQ) are justified by the additional risk taken when compared to the broader U.S. market benchmark, the S&P 500 (SPY). Historical price data was processed to compute return metrics, risk indicators, and performance consistency measures, which were then visualized in Power BI to support clear, decision-oriented insights.

The dashboard is structured into three analytical layers:

- The **Executive Overview** highlights long-term growth and high-level performance using CAGR and cumulative returns.
- The **Risk Analysis** section focuses on volatility, risk-adjusted returns measured by the Sharpe Ratio, and downside exposure through maximum drawdown.
- The **Performance Breakdown** examines return consistency using rolling returns, daily return distributions, and extreme monthly outcomes.

Key findings from the analysis show that QQQ delivers higher long-term returns than SPY but with increased volatility and deeper drawdowns. Despite this higher risk profile, QQQ demonstrates superior risk-adjusted performance, indicating that investors were compensated for the additional risk taken. Distribution and rolling return analysis further reveal that QQQ's outperformance is accompanied by greater variability, reinforcing the trade-off between growth potential and stability.

Overall, this project demonstrates the practical application of Power BI for financial analytics, emphasizing structured data modeling, meaningful KPI selection, and clear visual storytelling. The end-to-end workflow reflects real-world business intelligence practices and is suitable for inclusion as a portfolio project on GitHub and for discussion in professional data analyst interviews.

2. Business Problem & Analytical Objective

Equity investors and analysts are often faced with the decision of choosing between **high-growth strategies** and **broad market exposure**. QQQ, which tracks the NASDAQ-100, is widely perceived as a growth-oriented ETF with strong long-term return potential, while SPY, which tracks the S&P 500, is commonly used as a benchmark representing the overall U.S. equity market. While historical returns suggest that QQQ often outperforms SPY, this outperformance comes with higher perceived risk.

The core business problem addressed in this project is to determine whether **QQQ's higher returns are justified by the level of risk taken when compared to SPY**. From an investment and portfolio management perspective, raw returns alone are insufficient; decisions must account for volatility, downside risk, and return consistency over time.

The analytical objectives of this project are therefore to:

- Quantitatively compare the **long-term performance** of QQQ and SPY
- Evaluate and contrast their **risk characteristics**, including volatility and drawdowns
- Assess **risk-adjusted performance** using established financial metrics
- Provide clear, visual insights that enable faster interpretation by non-technical stakeholders

Power BI is used as the primary analytical interface to translate these objectives into an intuitive dashboard. By structuring the analysis into executive-level summaries, detailed risk evaluation, and performance consistency views, the dashboard is designed to support both **strategic decision-making** and **analytical deep dives**. This section establishes the foundation for the remainder of the report, framing the analysis not as a purely academic exercise, but as a **business-driven evaluation of risk versus reward** in equity investment choices.

3. Data Overview

This section describes the datasets used in the analysis, their scope, and the key fields that support the Power BI visuals and insights presented later in the report.

3.1 Overview of QQQ and SPY

The analysis focuses on two widely used U.S. equity ETFs:

- QQQ represents the NASDAQ-100 Index and provides exposure to large-cap, growth-oriented companies, with a strong concentration in the technology sector.
- SPY represents the S&P 500 Index and serves as a broad-market benchmark covering multiple sectors of the U.S. economy.
- Using these two ETFs enables a clear comparison between a growth-focused strategy and a diversified market portfolio.

3.2 Time Period and Data Granularity

The dataset consists of historical daily price data covering multiple market cycles, including periods of expansion, volatility, and drawdowns.

Daily granularity was intentionally chosen to allow:

- Accurate calculation of daily returns
- Reliable estimation of volatility and drawdowns
- Construction of rolling performance metrics
- This level of granularity supports both high-level performance evaluation and deeper risk analysis.

3.3 Key Data Fields Used for Analysis

The following fields form the foundation of the Power BI model and visuals:

- **Date** – Trading date used for time-series analysis and filtering
- **Adjusted Close Price** – Price adjusted for dividends and splits, used for return calculations
- **Daily Return** – Percentage change between consecutive trading days
- **Cumulative Return** – Growth of an investment over time
- **Ticker** – Identifier used to distinguish QQQ and SPY within a unified fact table

In addition to price-based data, a separate metrics dataset is used to store precomputed risk and performance indicators such as volatility, Sharpe ratio, maximum drawdown, beta, and alpha. These metrics are calculated in Python and imported into Power BI to ensure accuracy and reduce model complexity.

4. Tools & Technologies Used

This project combines data engineering, analytical computation, and visual analytics tools to deliver a structured and scalable financial analysis workflow. Each tool was selected based on its role in a real-world business intelligence environment.

Python – Data Engineering & Metric Computation

Python is used as the backend processing layer for the project. Its role is limited to data preparation and metric calculation, ensuring that Power BI focuses primarily on visualization and analysis rather than heavy computation.

Python is used to:

- Retrieve historical price data for QQQ and SPY
- Clean and standardize raw market data
- Calculate return-based metrics such as daily returns, cumulative returns, and rolling performance
- Compute risk and performance indicators including volatility, Sharpe ratio, maximum drawdown, beta, and alpha
- By performing these calculations upstream, the analytical logic remains transparent, reproducible, and easy to validate.

Power BI – Data Modeling, DAX, and Visualization

Power BI serves as the primary analytical and presentation layer of the project. Its strengths are leveraged in three key areas:

- Data modeling using a star schema to ensure consistent filtering and time-based analysis
- DAX measures for dynamic calculations such as CAGR and rolling returns
- Interactive visualizations that clearly communicate performance and risk trade-offs
- Power BI is intentionally used to explain insights visually, rather than to replicate complex financial formulas already computed in Python.

5. Data Modeling in Power BI

A well-structured data model is critical for accurate analysis and intuitive visualizations. In this project, Power BI uses a **star schema design** to ensure consistent filtering, efficient calculations, and scalability as additional metrics or datasets are added.

5.1 Star Schema Design

The data model is organized around a central fact table that stores time-series return data, supported by dimension tables that provide context. This approach avoids ambiguity in relationships and enables reliable time-based analysis across all visuals.

The star schema design was chosen because it:

- Simplifies relationships and reduces model complexity
- Ensures consistent behavior of slicers and filters
- Aligns with industry-standard business intelligence practices

5.2 Fact Returns Table

The **Fact Returns** table is the core of the model and contains daily return data for both ETFs. It includes:

- Date
- Adjusted Close Price
- Daily Return
- Cumulative Return
- Ticker (QQQ or SPY)

By combining QQQ and SPY into a single fact table and differentiating them using the Ticker field, Power BI visuals can easily compare both ETFs within the same chart using legends and filters.

5.3 Metrics Table

The **Metrics** table stores precomputed risk and performance indicators calculated in Python. These include:

- Annualized Volatility
- Sharpe Ratio
- Maximum Drawdown
- Beta (relative to SPY)
- Alpha

Separating these metrics into their own table avoids unnecessary recalculation in DAX and ensures that complex financial formulas remain transparent and auditable.

5.4 Date Dimension and Relationships

A dedicated **Date dimension** is used to manage all time-based analysis. It includes fields such as year, month, and year-month combinations, enabling flexible slicing and aggregation.

The relationship structure follows a one-to-many pattern:

- Date[Date] → Fact_Returns[Date]

This ensures that all visuals respond consistently to time-based slicers and prevents common issues caused by using multiple date fields across tables.

5.5 Rationale for the Chosen Model

This data modeling approach was selected to balance performance, clarity, and analytical flexibility. By isolating raw returns, risk metrics, and time attributes into appropriate tables, the model supports both executive-level summaries and detailed analytical views without redundancy or confusion.

The result is a clean, scalable Power BI model that serves as a strong foundation for the dashboard pages discussed in the following sections.

6. Dashboard Design Philosophy

Analytical Structure and Audience Focus:

The dashboard is designed to support both executive-level decision-making and analytical evaluation. To achieve this, the analysis is structured into three pages that progressively move from high-level performance insights to detailed risk and consistency assessment. Each page answers a specific business question, allowing users to quickly grasp key outcomes while retaining the ability to explore deeper patterns when required.

Metrics and visuals were selected to directly address the central objective of the project: evaluating the trade-off between higher returns and higher risk. Non-essential indicators were intentionally excluded to maintain clarity and analytical focus.

Visual Clarity and Interactivity:

Visual design emphasizes simplicity and interpretability. Consistent color usage is applied across all charts to distinguish QQQ and SPY, while minimal gridlines and clean layouts reduce visual noise. Interactivity is limited to time-based slicers, ensuring that all visuals remain synchronized without introducing unnecessary complexity.

These design choices ensure that insights are communicated clearly, enabling users to understand performance differences and risk exposure with minimal explanation.

7. Dashboard Page 1: Executive Overview

The Executive Overview page is designed to provide a **quick, high-level comparison of long-term performance** between QQQ and SPY. The layout prioritizes clarity, enabling stakeholders to understand growth outcomes and relative performance at a glance.

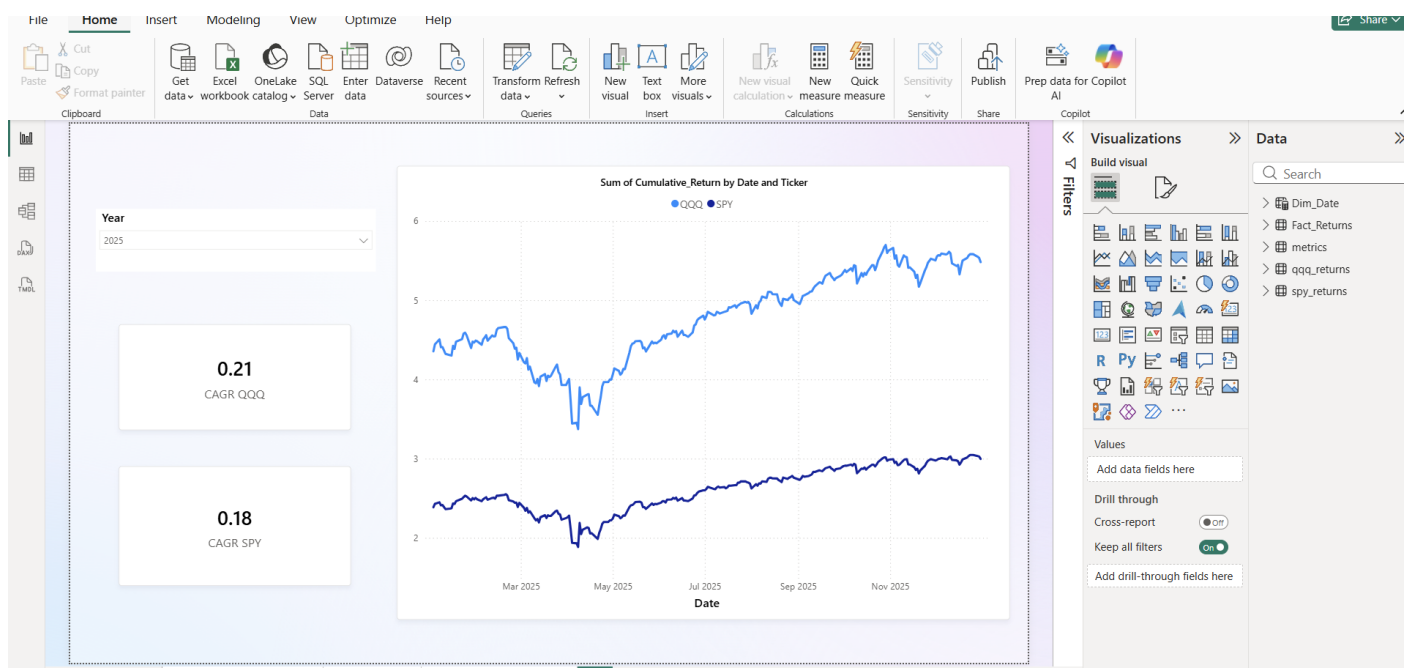


Figure 1 - Dashboard Page 1: Executive Overview

7.1 Year Slicer

The Year slicer is positioned at the **top-left corner** of the page to control the time context of the analysis. Selecting a year dynamically updates all visuals on the page, allowing users to focus on specific market periods.

This slicer enables targeted evaluation of performance during different phases such as market recoveries, corrections, or high-growth periods, while keeping the overall dashboard interaction simple.

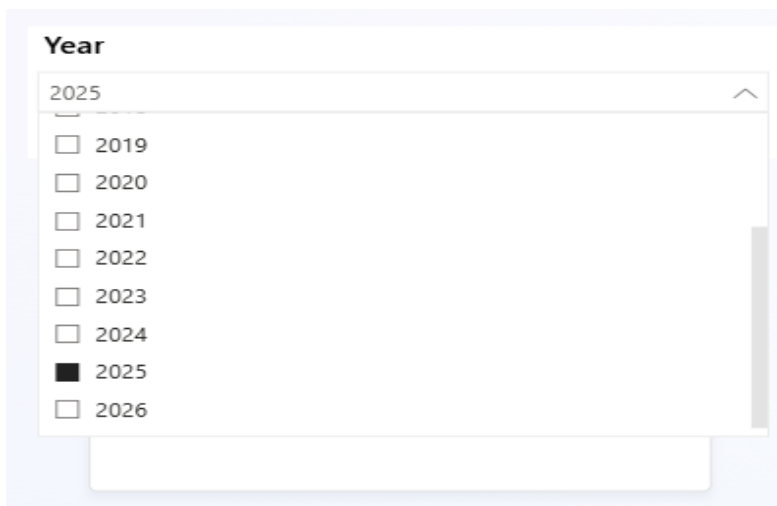


Figure 2 - Year Slicer

7.2 CAGR – QQQ

The **CAGR QQQ** KPI card is placed on the **left side of the dashboard**, directly below the Year slicer. This card displays the compound annual growth rate of QQQ for the selected time period.

This metric summarizes QQQ's long-term growth into a single annualized figure, answering the question: *What was the average annual growth delivered by QQQ over the selected period?*

The placement ensures that QQQ's growth performance is immediately visible before examining comparisons or trends.

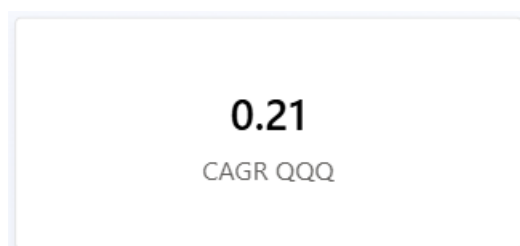


Figure 3 - CAGR – QQQ

7.3 CAGR – SPY

The **CAGR SPY** KPI card is positioned **below the QQQ CAGR card**, enabling a direct vertical comparison between the two ETFs.

SPY acts as the market benchmark in this analysis. Displaying both CAGR values together allows users to quickly assess whether QQQ's growth advantage over the broader market is meaningful within the selected timeframe.

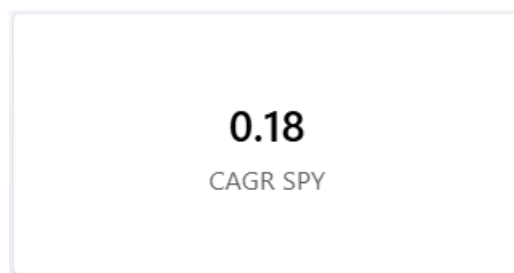


Figure 4 - CAGR – SPY

7.4 Cumulative Return Line Chart

The cumulative return line chart occupies the right side of the dashboard, taking up the largest visual area. This placement reflects its importance as the primary performance visualization.

The chart shows how cumulative returns for QQQ and SPY evolve over time, allowing users to observe:

- Performance divergence between the two ETFs
- Periods of drawdowns and recoveries
- The persistence of outperformance

Unlike CAGR, which is a summary metric, this chart reveals the path of returns, providing context around volatility, timing, and consistency.

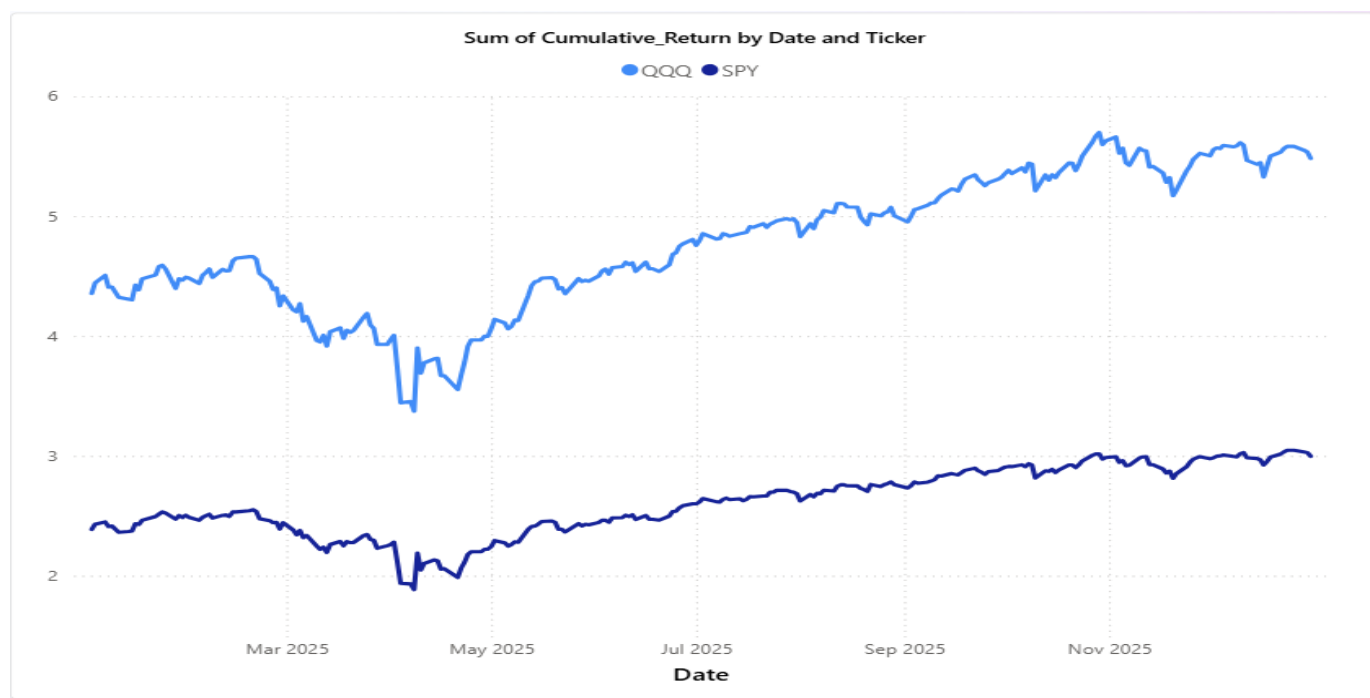


Figure 5 - Cumulative Return Line Chart

7.5 Key Insight from Page 1

Together, the KPI cards and cumulative return chart show that while both ETFs grow over time, QQQ demonstrates stronger long-term growth than SPY, as reflected in both CAGR and cumulative performance.

However, the fluctuations visible in the line chart indicate that this higher return is achieved with greater variability, setting the stage for the deeper risk analysis presented on the next page.

8. Dashboard Page 2: Risk Analysis

The Risk Analysis page evaluates the **risk profile behind the returns** observed on the Executive Overview page. While QQQ demonstrates stronger growth, this page answers whether that performance is achieved with acceptable risk relative to SPY.

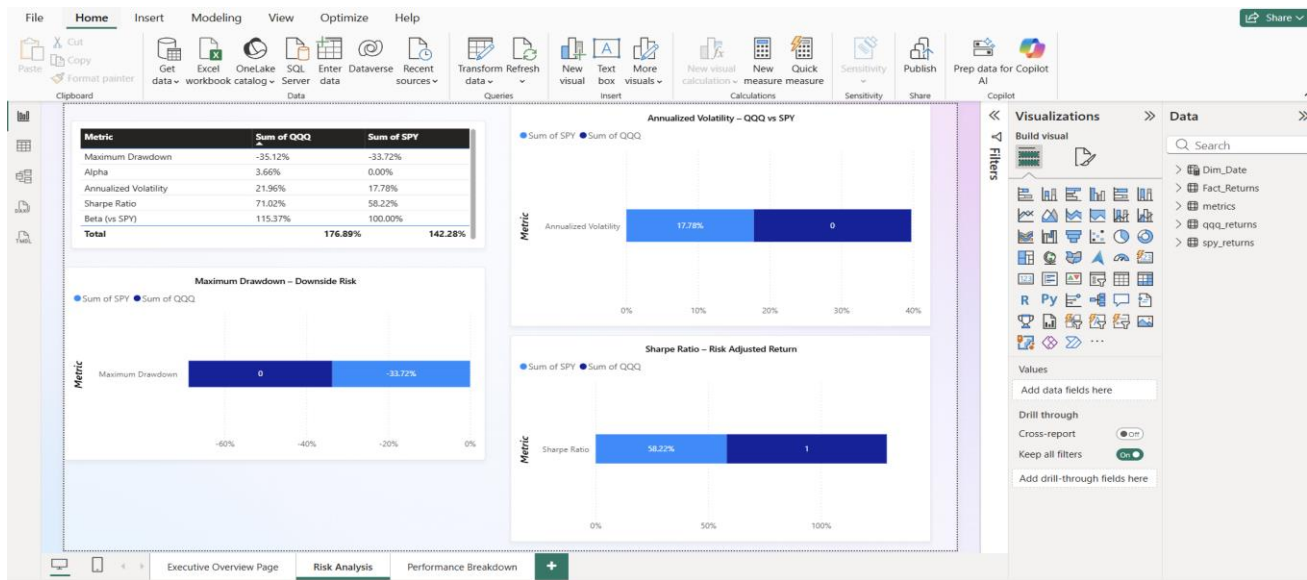


Figure 6 - Dashboard Page 2: Risk Analysis

8.1 Risk Metrics Summary Table

The risk metrics table is positioned in the **top-left section** of the page and acts as a consolidated reference for all key risk indicators.

The table contains:

- Maximum Drawdown
- Alpha
- Annualized Volatility
- Sharpe Ratio
- Beta (vs SPY)

This table enables side-by-side comparison of QQQ and SPY across multiple dimensions of risk and performance. It provides a factual baseline before interpreting the visual charts.

Metric	Sum of QQQ	Sum of SPY
Maximum Drawdown	-35.12%	-33.72%
Alpha	3.66%	0.00%
Annualized Volatility	21.96%	17.78%
Sharpe Ratio	71.02%	58.22%
Beta (vs SPY)	115.37%	100.00%
Total	176.89%	142.28%

Figure 7 - Risk Metrics Summary Tab

8.2 Annualized Volatility – QQQ vs SPY

The Annualized Volatility bar chart is placed in the **top-right section** to immediately highlight differences in overall risk magnitude.

Annualized volatility measures the degree of return fluctuation over time. Higher volatility indicates greater uncertainty and price swings.

This chart shows that **QQQ exhibits higher volatility than SPY**, confirming that QQQ's stronger returns are accompanied by increased variability.

This visual answers:

Which ETF experiences greater return instability?

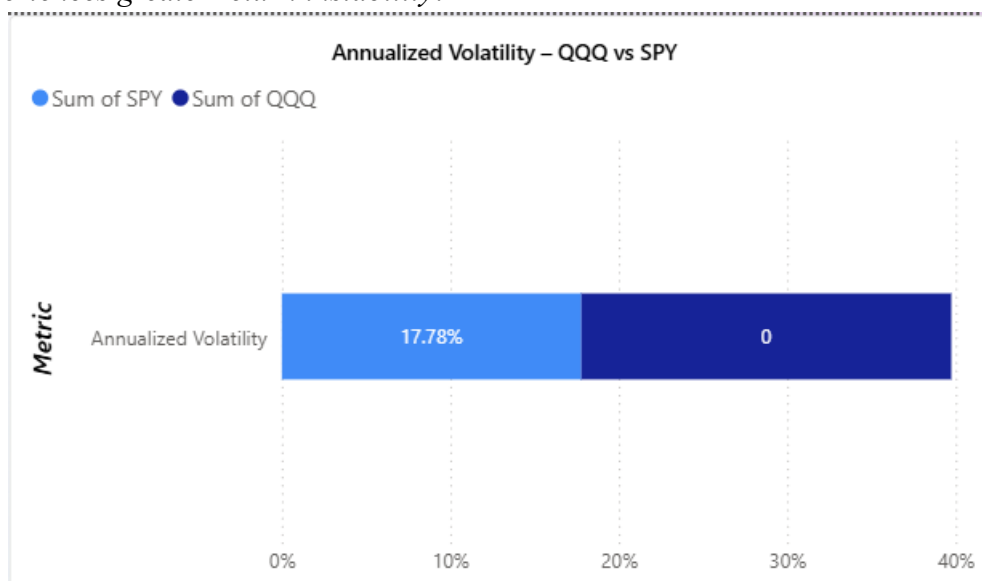


Figure 8 - Annualized Volatility – QQQ vs SPY

8.3 Maximum Drawdown – Downside Risk

The Maximum Drawdown chart is located in the **bottom-left section**, focusing specifically on downside risk. Maximum drawdown captures the **worst peak-to-trough loss** experienced during the analysis period. This metric is critical for understanding potential losses during market stress.

The chart illustrates that **QQQ experienced deeper drawdowns than SPY**, indicating higher exposure to severe losses during downturns.

This visual answers:

Which ETF exposes investors to larger worst-case losses?

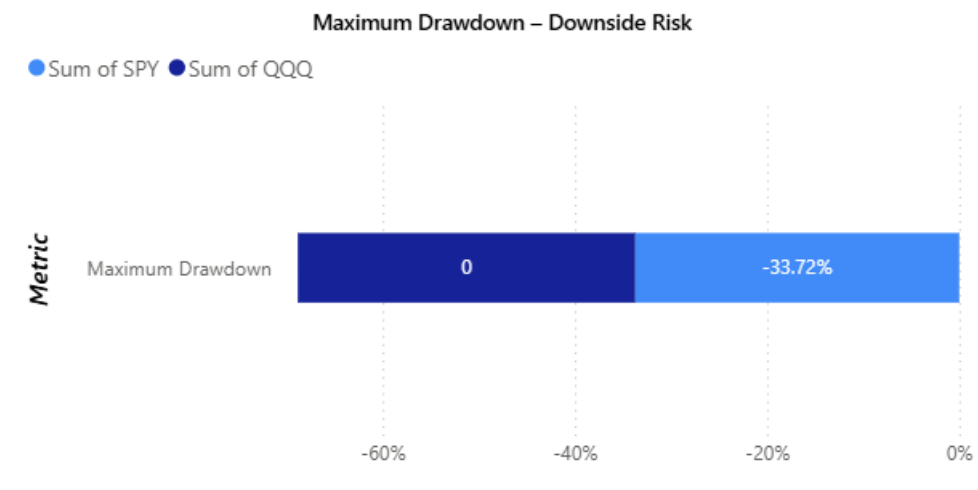


Figure 9 - Maximum Drawdown – Downside Risk

8.4 Sharpe Ratio – Risk Adjusted Return

The Sharpe Ratio chart is positioned in the **bottom-right section**, completing the risk narrative by shifting from raw risk to **risk-adjusted performance**.

The Sharpe Ratio measures how much return is earned per unit of risk. A higher value indicates more efficient risk-taking.

This chart shows that **QQQ achieves a higher Sharpe Ratio than SPY**, suggesting that despite higher volatility and drawdowns, QQQ compensates investors with superior risk-adjusted returns.

This visual answers:

Was QQQ's higher risk rewarded with proportionally higher returns?

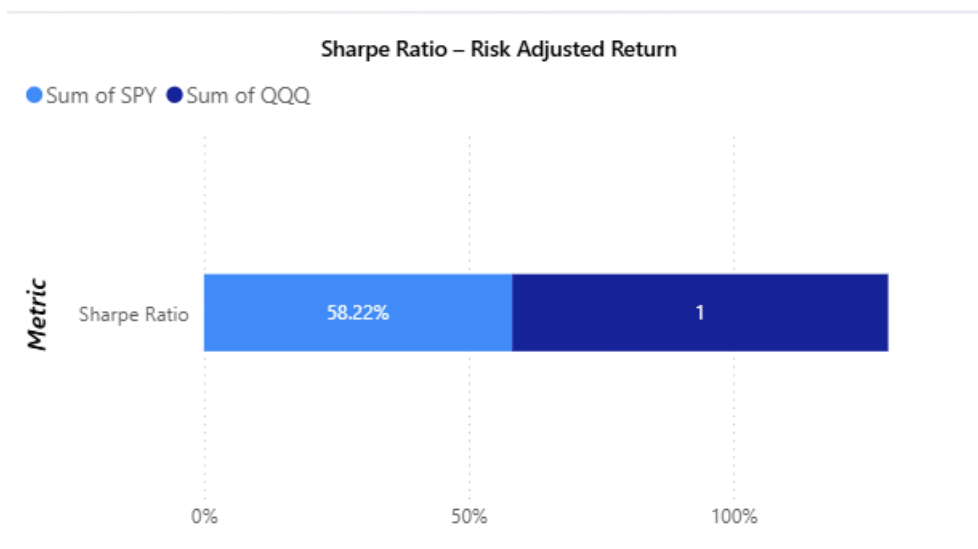


Figure 10 - Sharpe Ratio – Risk Adjusted Return

8.5 Key Insight from Risk Analysis

The Risk Analysis page demonstrates that QQQ carries **higher volatility and deeper drawdowns** than SPY. However, its **stronger Sharpe Ratio** indicates that the additional risk is efficiently compensated. This page

establishes the trade-off between growth and stability and sets the foundation for analyzing performance consistency and distribution on the next page.

9. Dashboard Page 3: Performance Breakdown

The Performance Breakdown page examines **return consistency and distribution**, moving beyond averages and risk summaries. This page focuses on *how returns behave over time* and *how extreme gains and losses are distributed* for QQQ and SPY.

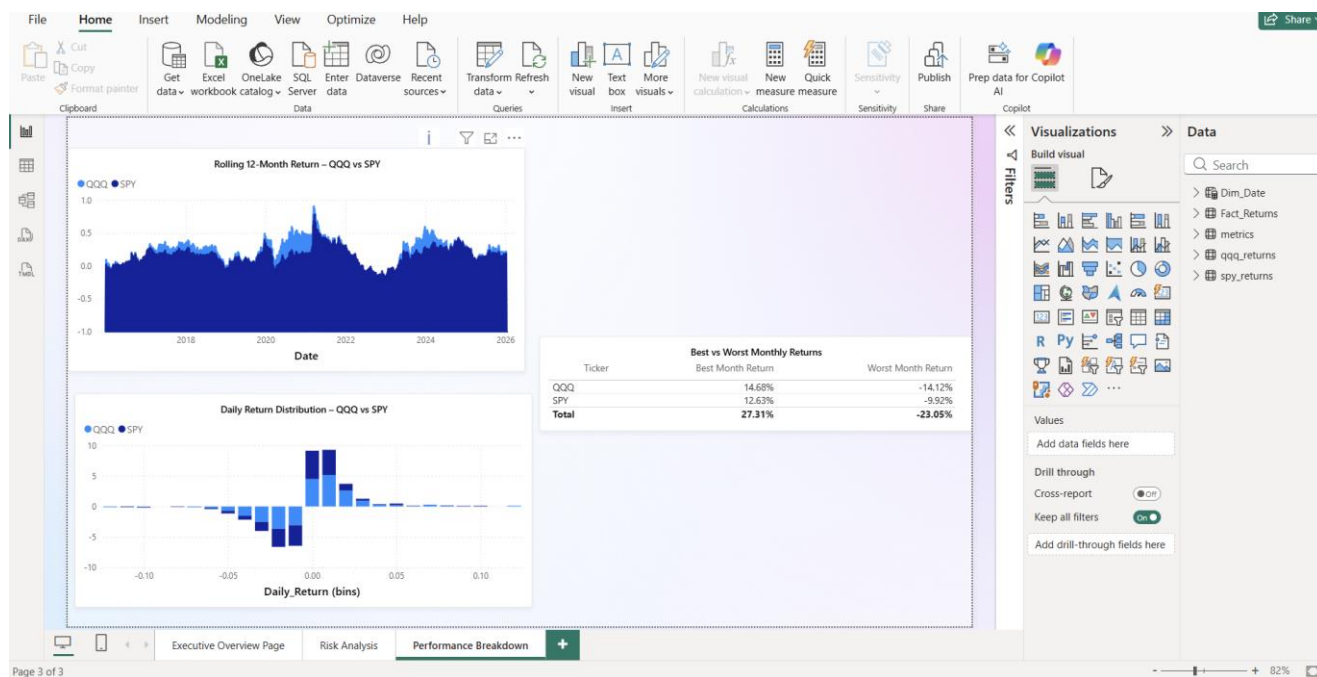


Figure 11 - Dashboard Page 3: Performance Breakdown

9.1 Rolling 12-Month Return – QQQ vs SPY

The Rolling 12-Month Return chart is placed in the **top-left section** of the page and is the primary visual for assessing return consistency.

This chart shows rolling one-year returns calculated at each point in time, smoothing short-term noise and highlighting medium-term performance cycles. Unlike cumulative returns, rolling returns reveal periods of sustained strength and weakness.

This visual helps identify:

- Periods where QQQ consistently outperformed SPY
- Cycles of underperformance or recovery
- Stability of returns across market regimes

The chart shows that **QQQ exhibits higher peaks and deeper troughs**, indicating stronger upside potential but greater variability compared to SPY.



Figure 12 - Rolling 12-Month Return – QQQ vs SPY

9.2 Daily Return Distribution – QQQ vs SPY

The daily return distribution histogram is positioned in the **bottom-left section**, focusing on short-term return behaviour.

This chart groups daily returns into bins and displays their frequency, allowing comparison of:

- Return concentration around zero
- Frequency of extreme positive or negative days
- Overall spread of daily returns

The wider distribution observed for QQQ indicates **greater daily volatility**, while SPY shows a more concentrated distribution, reflecting relative stability.

This visual answers:

How frequently do extreme daily moves occur for each ETF?

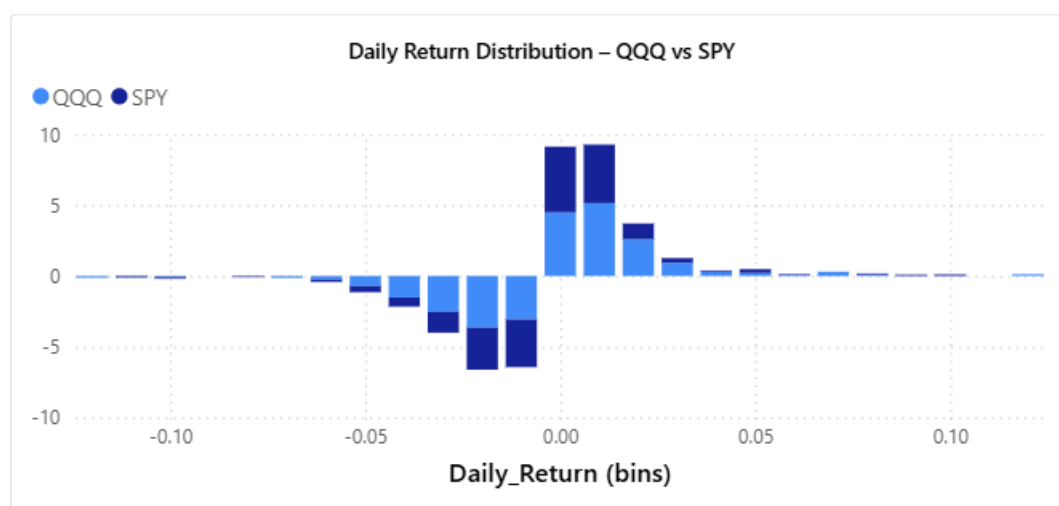


Figure 13 - Daily Return Distribution – QQQ vs SPY

9.3 Best vs Worst Monthly Returns Table

The Best vs Worst Monthly Returns table is placed on the **right side of the page**, summarizing extreme performance outcomes.

This table highlights:

- Best single-month return
- Worst single-month return
- Direct comparison between QQQ and SPY

The table shows that **QQQ achieves higher best-month returns but also suffers deeper worst-month losses**, reinforcing the high-risk, high-reward profile observed throughout the analysis.

This table provides concrete evidence of tail risk and upside potential that complements the distribution and rolling return charts.

Ticker	Best vs Worst Monthly Returns	
	Best Month Return	Worst Month Return
QQQ	14.68%	-14.12%
SPY	12.63%	-9.92%
Total	27.31%	-23.05%

Figure 14 - Best vs Worst Monthly Returns Table

9.4 Key Insights from Performance Breakdown

The Performance Breakdown page demonstrates that QQQ's superior long-term performance is driven by **stronger upside periods**, but these gains are accompanied by **greater return variability and deeper short-term losses**. SPY, while more stable, shows lower extremes on both the upside and downside.

This page completes the analytical narrative by validating that QQQ's higher CAGR and Sharpe Ratio are supported by observable return behavior, not just summary statistics.

10. Key Findings & Investment Interpretation

The analysis highlights a clear performance and risk differentiation between QQQ and SPY across all dashboard pages. QQQ consistently demonstrates **higher long-term growth**, as evidenced by superior CAGR and cumulative returns. This growth advantage is driven by strong upside periods, particularly visible in rolling 12-month return cycles. However, this performance comes with **higher volatility and deeper drawdowns**. QQQ experiences larger price swings and more severe peak-to-trough losses, increasing downside exposure during market stress.

Despite this higher risk, QQQ delivers **stronger risk-adjusted returns**, as shown by its higher Sharpe Ratio. This indicates that investors are compensated for the additional risk taken, making QQQ attractive for growth-oriented investors with higher risk tolerance. SPY, in contrast, offers **greater stability and lower downside risk**, making it more suitable for investors prioritizing capital preservation and smoother return profiles. From an investment perspective, QQQ aligns with **aggressive growth strategies**, while SPY serves as a **core market exposure** or defensive allocation.

11. Limitations & Assumptions

Several assumptions and limitations should be considered when interpreting the results. The analysis relies on **historical price data**, which does not guarantee future performance. Market dynamics, macroeconomic conditions, and structural changes may alter return and risk behavior going forward. Transaction costs, taxes, and tracking errors are not included, which may impact real-world investment outcomes. Risk-free rates used

for Sharpe Ratio calculations are assumed to be constant, which may differ from actual market conditions over time. Finally, the analysis focuses on **two ETFs only**, and results should not be generalized to other asset classes without further study.

12. Conclusion

This project demonstrates an end-to-end financial analytics workflow using **Python for data processing** and **Power BI for professional-grade visualization and analysis**. The dashboards clearly show that QQQ delivers higher returns at the cost of increased risk, while SPY provides more stable but lower growth. By combining performance, risk, and distribution-based analysis, the project offers a comprehensive view of the return-risk trade-off.

From a business intelligence perspective, the project highlights strong data modeling, effective use of DAX, and purposeful dashboard design. The structure and insights presented are suitable for both analytical roles and stakeholder-facing reporting, making this project a strong portfolio asset for a data analyst role.